

SARANYA VENKATRAMAN

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EDUCATION

The Pennsylvania State University , University Park, PA, USA MS (cont.)/PhD- PIKE Lab College of Information Science and Technology (IST)	<i>Aug 2018–Dec 2024</i>
New York University , New York City, NY, USA Visiting Scholar - ML² Lab Courant Institute of Mathematical Sciences (CIMS)	<i>Jan 2020–Jan 2024</i>
GB Pant Engineering College , New Delhi, India BTech in Computer Science & Engineering	<i>Aug 2012–July 2016</i>

RESEARCH INTERESTS

My research lies at the intersection of Large Language Models, Multi-Agent Systems, and Computational Cognitive Modeling, with a focus on building agents that simulate realistic human behavior and decision-making. Broadly, I am interested in advancing AI systems and Reinforcement Learning methods that are not only capable but also interpretable and behaviorally faithful.

WORK EXPERIENCE

Amazon, Santa Clara, CA <i>Applied Scientist - Amazon Stores</i>	Feb 2025–Present
· Research on agentic user simulation, building LLM-powered digital twins of customers that model joint decision-making to evaluate shopping experiences at scale. · Build scalable pipelines for persona generation and query synthesis, and evaluate agent reasoning and behavioral fidelity against ground-truth customer signals. · Partner with engineering, science, product, and legal teams to bring agentic simulation to products across Amazon.	
Google, New York, NY <i>Research Intern - Google Assistant</i>	May–Sept 2020
· Built a hierarchical Soft Actor-Critic (RL) recommender dialog agent for a hybrid action space that interprets user preferences and suggests items from a marketplace using TF-Agents. · Conducted ablation studies with agent actions, observations and data simulation techniques to populate a continuous space marketplace using Google Vizier for hyperparameter tuning.	
Samsung Research America (SRA), Mountain View, CA <i>Intern - Artificial Intelligence Center</i>	May–Aug 2018
· Developed a rule-based components extraction algorithm based on nested parsing of constituency-based syntax trees derived from natural language utterances. · Implemented a semantic search based intent-to-action mapper using an ensemble of a short sentence similarity service and GloVe word embeddings. Deployed as a web service (REST API) and integrated end-to-end with mobile devices using Bixby Capsule SDK.	

Cadence Design Systems, San Jose, CA

May–Aug 2017

Machine Learning Intern - Machine Learning Team (now MAGESTIC), R&D Center

- Set up feature extraction and machine-learning pipelines for Cadence’s proprietary layout images.
- Developed a proof of concept for hierarchical clustering & assisted-predictive labeling on a custom-built dataset.
- Achieved an accuracy of 75% for 4 levels of abstraction by using an ensemble of wavelet transformations with density-based clustering approaches.

SELECTED PROJECTS

AgentTwin: Agentic Twins of Mobile Users

Keywords: Agentic AI, User Simulation, Behavioral Fidelity, Persona Generation

- At Amazon Stores, I worked on AgentTwin, an end-to-end framework for LLM-powered agentic twins that model how real users make decisions on mobile shopping apps. AgentTwin combines persona generation, self-evolving memory with reflection, and a planner–critic–operator agent for high-fidelity device control, with evaluation against ground-truth behavioral signals and automated reasoning analysis.

GPT-who: Psycholinguistic Signals for LLM Authorship Detection

Keywords: Authorship Attribution, AI Text Detection, Uniform Information Density, Interpretability

- During my PhD dissertation work, I developed GPT-Who, a psycholinguistically inspired domain-agnostic detector for distinguishing LLM-generated from human-generated text, leveraging Uniform Information Density (UID) theory to model the unique statistical signature of each author.
- GPT-Who outperformed state-of-the-art detectors (GLTR, GPTZero, DetectGPT, OpenAI detector) by over 20% across domains while remaining computationally efficient and interpretable.

TECHNICAL SKILLS

Computer Languages Packages & Tools

Python, JavaScript, C/C++, MATLAB, R
PyTorch, Transformers, TensorFlow, TF-Agents, Keras, Flask,
SciPy, Scikit-learn, Pandas, NumPy, Firebase

PUBLICATIONS

Saranya Venkatraman, Thanh Tran, Shunyan Luo, Devin Chen, Yuning Wu, Kai Wei, Allie Colin, Guido Imbens, Ido Rosen, and Christine Ambrose. “[AgentTwin: An End-to-End Framework for Building Agentic Twins of Mobile Users](#).” COLM 2026 (Workshop on Agent Behavior).

Saranya Venkatraman*, Thanh Tran*, Shunyan Luo, Devin Chen, Yuning Wu, Kai Wei. “[MobiTwin: Toward Building End-to-End Framework for User’s Mobile Twins](#).” IEEE CogMI 2025 (Agentic Intelligence: Risks, Ethics, and Trust Workshop).

Nafis Irtiza Tripto, **Saranya Venkatraman**, Mahjabin Nahar, and Dongwon Lee. “[Beyond Checkmate: Exploring the Creative Choke Points for AI Generated Texts](#).” EMNLP 2025.

Ekaterina Artemova, Jason S. Lucas, **Saranya Venkatraman**, et al. “[Beemo: Benchmark of Expert-edited Machine-generated Outputs](#).” NAACL 2025.

Saranya Venkatraman, Nafis Irtiza Tripto, and Dongwon Lee. “[CollabStory: Multi-LLM Collaborative Story Generation and Authorship Analysis](#).” NAACL Findings 2025.

Saranya Venkatraman, Adaku Uchendu, and Dongwon Lee. “[GPT-who: An Information Density-based Machine-Generated Text Detector](#).” NAACL Findings 2024.

Nafis Irtiza Tripto, **Saranya Venkatraman**, Dominik Macko, Robert Moro, Ivan Srba, Adaku Uchendu, Thai Le, and Dongwon Lee. “[A Ship of Theseus: Curious Cases of Paraphrasing in LLM-Generated Texts](#).” ACL 2024.

Eric Xing, **Saranya Venkatraman**, Thai Le, and Dongwon Lee. “[ALISON: Fast and Effective Stylometric Authorship Obfuscation](#).” AAAI 2024.

Tricia J. Ngoon, Sushil S., Angela E.B. Stewart, Ung-Sang Lee, **Saranya Venkatraman**, Neil Thawani, Prasenjit Mitra, Sherice Clarke, John Zimmerman, and Amy Ogan. “[ClassInSight: Designing Conversation Support Tools to Visualize Classroom Discussion for Personalized Teacher Professional Development](#).” CHI 2024.

Pranav Venkit, Mukund Srinath, Sanjana Gautam, **Saranya Venkatraman**, Vipul Gupta, Rebecca J. Passonneau, and Shomir Wilson. “[The Sentiment Problem: A Critical Survey towards Deconstructing Sentiment Analysis](#).” EMNLP 2023.

Saranya Venkatraman, He He, and David Reitter. “[How do decoding algorithms distribute information in dialogue responses?](#)” EACL Findings 2023.

Saranya Venkatraman, Prasenjit Mitra, Sherice N. Clarke, Andrea Gomoll, Zaynab Gates, Sushil S., Tarang Tripathi, and Amy Ogan. “[ClassInSight: Automating Analysis of Classroom Discussions to Support Teacher Noticing and Reflection on Dialogic Pedagogy](#).” EARLI 2021.

Greeshma Sharma, **Saranya Venkatraman**, and Virender Singh. “[Ambiguity in Semantic Integration: A decomposition analysis by ERP](#).” International Journal of Engineering Sciences & Technology 2277-9655 (2016).

Greeshma Sharma, **Saranya Venkatraman**, et al. “[Artificial Neural Network in Virtual Reality: A Survey](#).” International Journal of Virtual Reality. 15 (02): 44-52 (2015).

TUTORIALS

Adaku Uchendu, **Saranya Venkatraman**, Thai Le, Dongwon Lee. “[Catch Me If You GPT: Tutorial on Deepfake Texts](#).” NAACL 2024.

Adaku Uchendu, Vladislav Mikhailov, Jooyoung Lee, **Saranya Venkatraman** et al. “[Tutorial on Artificial Text Detection](#).” INLG 2022.

SERVICE & RECOGNITION

Amazon Inventor Award & Earn Trust Award (Leadership Principle Recognition)	2025–2026
Graduate Student Award for Excellence in Teaching Support	2023–2024
Outstanding Paper Award at EMNLP (Venkit et al., 2023)	2023
Scholarly Reviewer for ACL, NAACL, EMNLP, AAAI	2017–2026
Tutor & Mentor for NSF Research Experiences for Undergraduates (REU)	2023–2024
Secretary for Graduates in IST (GIST), Graduate Student Organization	2020–2021